

Supplementary Materials

A Geometric-Functional Consistency Framework for Evaluating Protein Interaction Network Embeddings

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Contents:
Table S1: G-F curve statistics across 10 random subsets (extended to $r = 0.55$)
Table S2: Purity comparison between DM and PCA embeddings
Algorithm S1: G-F Curve Construction

Table S1. G-F curve statistics across 10 random subsets (mean $\pm$ SD).

r	DM Purity	MDS Purity	DM Modularity	MDS Modularity
0.050	0.9111 $\pm$ 0.0091	0.9136 $\pm$ 0.0168	0.8974 $\pm$ 0.0106	0.8487 $\pm$ 0.0125
0.067	0.8655 $\pm$ 0.0162	0.8686 $\pm$ 0.0158	0.8545 $\pm$ 0.0139	0.8186 $\pm$ 0.0214
0.084	0.8162 $\pm$ 0.0210	0.8387 $\pm$ 0.0232	0.8185 $\pm$ 0.0119	0.7684 $\pm$ 0.0178
0.102	0.7573 $\pm$ 0.0237	0.8264 $\pm$ 0.0297	0.7719 $\pm$ 0.0077	0.6900 $\pm$ 0.0235
0.119	0.7271 $\pm$ 0.0350	0.7916 $\pm$ 0.0277	0.7263 $\pm$ 0.0158	0.6616 $\pm$ 0.0489
0.136	0.7169 $\pm$ 0.0167	0.7566 $\pm$ 0.0232	0.6851 $\pm$ 0.0209	0.5930 $\pm$ 0.0312
0.153	0.6820 $\pm$ 0.0245	0.7368 $\pm$ 0.0343	0.6386 $\pm$ 0.0192	0.5770 $\pm$ 0.0338
0.171	0.6620 $\pm$ 0.0412	0.7197 $\pm$ 0.0327	0.5969 $\pm$ 0.0174	0.5384 $\pm$ 0.0249
0.188	0.6306 $\pm$ 0.0600	0.6996 $\pm$ 0.0276	0.5772 $\pm$ 0.0099	0.4854 $\pm$ 0.0340
0.205	0.5972 $\pm$ 0.0585	0.6088 $\pm$ 0.0958	0.5385 $\pm$ 0.0199	0.4311 $\pm$ 0.0337

r	DM Purity	MDS Purity	DM Modularity	MDS Modularity
0.222	0.5500 ± 0.0451	0.4761 ± 0.0856	0.4995 ± 0.0157	0.3751 ± 0.0363
0.240	0.5523 ± 0.0538	0.4565 ± 0.0496	0.4954 ± 0.0066	0.3835 ± 0.0324
0.257	0.5224 ± 0.0418	0.5153 ± 0.0639	0.4752 ± 0.0093	0.3617 ± 0.0209
0.274	0.5306 ± 0.0222	0.4975 ± 0.0278	0.4463 ± 0.0191	0.3422 ± 0.0128
0.291	0.5229 ± 0.0196	0.4691 ± 0.0587	0.4198 ± 0.0229	0.3371 ± 0.0116
0.309	0.5305 ± 0.0131	0.4252 ± 0.0850	0.3907 ± 0.0226	0.3129 ± 0.0189
0.326	0.5344 ± 0.0040	0.3879 ± 0.0769	0.3710 ± 0.0168	0.2944 ± 0.0074
0.343	0.5324 ± 0.0043	0.3490 ± 0.0585	0.3603 ± 0.0143	0.2833 ± 0.0049
0.360	0.5322 ± 0.0042	0.4466 ± 0.0565	0.3437 ± 0.0048	0.2774 ± 0.0073
0.378	0.5314 ± 0.0026	0.3860 ± 0.0773	0.3376 ± 0.0062	0.2716 ± 0.0068

r	DM Purity	MDS Purity	DM Modularity	MDS Modularity
0.395	0.5330 ± 0.0045	0.3267 ± 0.0401	0.3269 ± 0.0058	0.2698 ± 0.0064
0.412	0.5350 ± 0.0052	0.3120 ± 0.0051	0.3146 ± 0.0074	0.2671 ± 0.0068
0.429	0.4647 ± 0.1052	0.3116 ± 0.0056	0.3036 ± 0.0053	0.2606 ± 0.0068
0.447	0.4649 ± 0.1051	0.3029 ± 0.0068	0.2910 ± 0.0051	0.2575 ± 0.0065
0.464	0.3966 ± 0.0978	0.3008 ± 0.0059	0.2803 ± 0.0047	0.2536 ± 0.0066
0.481	0.3553 ± 0.0410	0.2967 ± 0.0048	0.2684 ± 0.0038	0.2488 ± 0.0066
0.498	0.3478 ± 0.0405	0.3011 ± 0.0064	0.2557 ± 0.0033	0.2424 ± 0.0065
0.516	0.3135 ± 0.0239	0.3082 ± 0.0081	0.2440 ± 0.0031	0.2362 ± 0.0067
0.533	0.3048 ± 0.0069	0.3057 ± 0.0066	0.2310 ± 0.0028	0.2276 ± 0.0073
0.550	0.3037 ± 0.0052	0.3041 ± 0.0063	0.2184 ± 0.0024	0.2204 ± 0.0071

Note: Data shown for all 30 r values. Full data available in the code repository.

Table S2. Purity comparison between DM and PCA embeddings.

r	DM Purity	PCA Purity
0.050	0.9174	0.8851
0.066	0.8870	0.8594
0.081	0.8563	0.7683
0.097	0.7660	0.7414
0.112	0.7468	0.7134
0.128	0.7126	0.5968
0.143	0.6725	0.5576
0.159	0.6793	0.5469
0.174	0.6432	0.5595
0.190	0.5851	0.5320
0.205	0.6059	0.5488
0.221	0.5073	0.5615
0.236	0.4812	0.5287
0.252	0.5100	0.5399
0.267	0.5340	0.5649
0.283	0.5159	0.5295
0.298	0.5307	0.5264
0.314	0.5343	0.5264
0.329	0.5310	0.5427
0.345	0.5308	0.5232
0.360	0.5308	0.5310
0.376	0.5277	0.5276
0.391	0.5277	0.5441
0.407	0.5308	0.3189
0.422	0.5312	0.3189

r	DM Purity	PCA Purity
0.438	0.5319	0.3235
0.453	0.3029	0.3212
0.469	0.3773	0.3212
0.484	0.3752	0.3235
0.500	0.3140	0.3235

Algorithm S1: G-F Curve Construction

Input:

- Embedding coordinates for N nodes
- GO annotation mapping for each node
- Range of distance thresholds $r = [0.05, 0.55]$ with step Δr
- Community detection algorithm (greedy modularity optimization)

Output:

- G-F curves: $\text{purity}(r)$, $\text{modularity}(r)$

Procedure:

1. Compute pairwise Euclidean distance matrix D ($N \times N$)
2. For each r in r_range :
 - a. Construct graph $G(r)$ by adding edge (i,j) if $D[i,j] < r$ and $i \neq j$
 - b. Apply greedy modularity optimization to $G(r)$ to obtain communities
 - c. Compute functional purity:
 $\text{purity} = \text{mean over communities of } (\text{max GO term count} / \text{community size})$
 - d. Compute Newman-Girvan modularity $Q(r)$
3. Plot $\text{purity}(r)$ and $Q(r)$ against r
4. Return curves

Notes:

- Node ordering is fixed before community detection for reproducibility.
- All random seeds are fixed (`random.seed(42)`, `numpy.random.seed(42)`).
- Complete implementation at <https://github.com/Qinrayn/GF-consistency-framework>